

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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(Dated: June 19, 2026)

Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects need not add. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with a synergy order parameter $\mathcal{S}$ (meV/atom). PBE+DFT-D3 on tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) gives opposite strain coefficients for N (-303 meV/%) and B ($+57$ meV/%) and $|\mathcal{S}|$ up to ~ 32 meV/atom at $+3\%$ strain, largest for P despite small $|\alpha|$. The pattern warns that sequential high-throughput screens can mis-rank stability when local dopant distortion couples nonlinearly to external strain.

INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1, 2]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications, yet charge transport in molecular networks is often polaronic [3, 4]. Quasi-hexagonal (qHP) C₆₀ graphullerene [3, 4] provides a covalently linked model network for such coupled tuning.

Beyond pristine monolayers, strain and substitution independently tune qHP C₆₀ band edges [2, 5, 6]. Non-additive strain–doping coupling has been reported in other two-dimensional networks [1]; a quantitative energy cross term $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a common PBE+D3 protocol. If formation energies or strain penalties are estimated from *separate* doping-only and strain-only scans, the omitted cross term can reach tens of meV/atom in coupled cells (this work)—large enough to mis-rank dopant stability or strain set-points in screening workflows.

This work establishes the energy coupling metric $\mathcal{S}$ on shared tetramer and size-scaling grids; Marcus λ and hopping observables are deferred to Supplemental Material.

COMPUTATIONAL METHODS

Setup and grids

We use CP2K with PBE+DFT-D3 on shared tetramer and $n \times \text{C}_{60}$ supercell grids ($\gtrsim 90$ converged tasks; Supplemental Material Table S2), deliberately below the Koopmans/rVV10 level of Ref. 3 to enable uniform (ϵ, δ) scans. Biaxial strain scales the pristine in-plane lattice at fixed fractional coordinates (rigid-strain upper bound; ionic-relaxation benchmark Table S3 pending). Tetramers

carry $\sim 5\%$ B/N/P substitution (seed 42); periodic cells carry one dopant per C₆₀ ($n = 1\text{--}8$, 280 Ry for $n \geq 6$ vs. 300 Ry below). Full basis, SCF, and grid definitions are in Supplemental Material.

Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is reported relative to pristine tetramers at the same strain (no chemical-potential reference). In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_f(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the substitution-energy scale: at $n = 4$ and $+3\%$ strain, P has $|\mathcal{S}| \approx 24$ meV/atom while the per-atom substitution scale is $|E_f| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_f| \approx 16$ eV per P site, $n_{\text{dop}}=12$, Table S1)—i.e. $|\mathcal{S}|/|E_f|_{\text{per atom}} \sim 3\%$, large enough to reorder stability margins in sequential doping-then-strain screens even though $|\mathcal{S}|$ exceeds the rigid per-atom strain increment ($\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$ meV/atom for P).

Non-additive coupling metric. We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond

the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

To first order in strain, $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$ when $\alpha_\delta = dE/d\epsilon$ is evaluated for the doped cell and α_0 for the pristine reference at the same n (rigid-strain protocol); this links the mechanical coefficient α to the cross term at small ϵ .

Periodic-cell $\mathcal{S}$ values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Supplemental Material Fig. S6). Following Capobianco *et al.* [3], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8), $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$, on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75\text{--}81$ meV reported for pristine qHP networks [3], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Supplemental Material). Marcus reorganization energies follow the vertical–adiabatic decomposition [3]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5.

RESULTS

Results focus on total-energy non-additivity: band-edge trends provide context, α sets the linear strain scale, and $\mathcal{S}$ quantifies the omitted cross term (IPR/ J and Marcus λ in Supplemental Material Fig. S5–S6 only).

Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from tetramer single-points at $\epsilon \in \{-5, 0, +5\}\%$. B introduces hole states near the valence manifold (acceptor character), compressing the π gap toward zero at $\epsilon = 0$, whereas N donates electrons into the conduction manifold (donor character) and P (larger covalent radius) perturbs the network more heterogeneously—hence the dopant-specific gap slopes in panel (a). Figure 1(b) shows π -DOS for B @ $\epsilon = 0$.

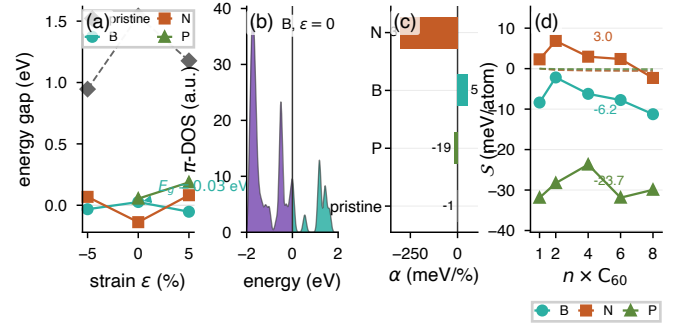


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO–LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon = 0$ (Electron SI palette). (c) Mechanical α . (d) $\mathcal{S}(n)$ at +3%; inset: schematic IPR/ J grid (details in Supplemental Material Fig. S6).

Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens a wide spread: N ($\alpha = -303.2$ meV/%, tension stabilizes), B (+56.9 meV/%, tension destabilizes), P (−19.1 meV/%) (Table S1). Figure 1(c) highlights the opposite α for N vs. B; ranking dopants by $\epsilon = 0$ stability alone is misleading.

Non-additive strain–doping coupling

Figure 1(d) tracks $\mathcal{S}(n)$ with one dopant per C_{60} . At $n = 4$ and +3% strain, $\mathcal{S} = -6.2$ meV/atom (B), +3.0 meV/atom (N), and −23.7 meV/atom (P); P shows the largest magnitude. Fifteen coupled points span $n \in \{1, 2, 4, 6, 8\}$ and B/N/P at +3%. For N, $\mathcal{S}$ changes sign between $n = 6$ (+2.4 meV/atom) and $n = 8$ (−2.3 meV/atom). Because $n = 6\text{--}8$ use 280 Ry whereas $n \leq 5$ use 300 Ry, this reversal may partly reflect basis-set change rather than intrinsic size physics; we treat it as provisional pending a matched-cutoff $n = 6$ benchmark.

DISCUSSION

Context. Prior work on pristine qHP C_{60} [3] and doped graphullerene networks [1, 2, 5] motivates coupled (ϵ, δ) scans; here we isolate the *total-energy* cross term $\mathcal{S}$, distinct from hopping metrics in Supplemental Material Fig. S6.

Mechanistic synthesis (three coupling classes).

(i) Opposite α for N vs. B. N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α)—consistent with donor vs. acceptor π -counting and the PDOS shifts in Fig. 1(a,b). **(ii) Large $|\mathcal{S}|$ for**

P despite small $|\alpha|$. P’s covalent radius (~ 107 pm vs. ~ 77 pm for C) and longer P–C bonds impose intrinsic cage distortion; rigid external strain couples to this built-in local stress, yielding a cross term that is invisible in separate scans but large when strain and substitutional doping are applied jointly—consistent with P’s intermediate E_f yet largest $|\mathcal{S}|$ at +3%. **(iii) N $\mathcal{S}$ sign change with n .** With one dopant per C_{60} , electrostatic and strain fields are dilute at large n ; the sign flip between $n = 6$ and $n = 8$ may reflect finite-size electrostatics but is confounded with the 280 Ry cutoff introduced at $n \geq 6$ —interpretation is deferred until a constant-cutoff control. Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n).

(iv) Linear $\mathcal{S}$ – α check. Using tetramer α (Table S1) in $\mathcal{S} \approx (\alpha_\delta - \alpha_0)\epsilon$ at $\epsilon = +3\%$ and dividing by 240 sites gives $\sim +0.7$, -3.8 , and -0.2 meV/atom for B/N/P—far smaller than the periodic $n=4$ values (-6.2 , $+3.0$, -23.7 meV/atom). The gap shows that tetramer-scale linear mechanics and 5% doping do not quantitatively predict periodic $\mathcal{S}$; for P, nonlinear coupling dominates beyond first-order $(\alpha_\delta - \alpha_0)\epsilon$.

(v) Significance versus DFT uncertainty. At $n=4$ and +3% strain, $|\mathcal{S}| \sim 24$ meV/atom is $\sim 3\%$ of the per-atom substitution-energy scale (Table S1) but still $\gg$ typical rigid per-atom strain increments from α . PBE formation energies for molecular solids can carry $\mathcal{O}(10^2)$ meV/atom uncertainties; we therefore treat $\mathcal{S}$ as a *coupled-scan correction* to sequential workflows rather than an absolute formation-energy prediction, and flag cases where $|\mathcal{S}|$ exceeds the rigid strain increment as the primary screening hazard.

Design implications. At +3% strain, P carries the largest additive-approximation error in $\mathcal{S}$ ($|\mathcal{S}| \sim 24$ – 32 meV/atom); co-design should treat P under tension as a *coupled* stability problem, not $|E_f|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. Gap closing [Fig. 1(a)] does not track

$|\mathcal{S}|$; we do not compute $\mathcal{S}_{\text{gap}}$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone. N $\mathcal{S}$ sign reversal by $n = 8$ cautions against tetramer-only extrapolation.

Limitations and outlook. Rigid strain (upper-bound on $\mathcal{S}$; ionic-relaxation check Table S3 pending); fixed-seed substitutional sites without thermodynamic ranking; mixed 280/300 Ry cutoffs across n (N $\mathcal{S}$ sign change at $n = 6$ – 8); tetramer vs. scaling concentration mismatch; PBE+D3 without Koopmans/rVV10; gap non-additivity ($\mathcal{S}_{\text{gap}}$) not quantified; Marcus λ and vertical transitions from charged geometry optimization still pending (Supplemental Material).

CONCLUSIONS

Using qHP C_{60} as a model covalent molecular network, we quantified non-additive strain–doping coupling in total energy. Opposite α for N vs. B and finite $\mathcal{S}$ up to ~ 32 meV/atom show that joint (ϵ, δ) tuning is not equivalent to sequential strain and doping scans; for P under tension, the cross term is a few percent of the per-atom substitution-energy scale yet can significantly alter stability rankings in sequential screening workflows. These energy-landscape metrics complement—but do not replace—Marcus λ from charged-state vertical transitions needed to connect $\mathcal{S}$ to polaron hopping rates.

We acknowledge computational resources and support from Xiamen University.

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